SENSE-MINE

AI-Enabled Wireless Surface Mesh Network for Real-Time Mine Subsidence Monitoring, Prediction and Early Warning

Detailed Project Report (DPR)

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Team Name: [Team Name]
Team Members: [Name 1, Name 2, Name 3, Name 4, Name 5, Name 6]
Institution: [Institution Name]

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1. Introduction

1.1 Background

Underground coal mining alters the stress distribution of overlying strata, and over time this can propagate to the surface as subsidence — a slow or sudden lowering of ground level that manifests as cracking, tilting, differential settlement, and in severe cases sinkholes. Surface subsidence over active or abandoned underground coal panels is a recognized operational safety concern in Indian coalfields, and periodic surface surveys, crack/depression monitoring and subsidence monitoring stations are already part of standard monitoring practice, as reflected in the Directorate General of Mines Safety (DGMS) annual reporting. Existing monitoring approaches — conventional leveling/GNSS surveys, total-station measurement, and satellite-based InSAR — each have real strengths but also real gaps for continuous, localized, low-cost surface monitoring: leveling and GNSS are accurate but labor-intensive, periodic and spatially sparse; InSAR gives excellent spatial coverage but is line-of-sight limited, has a multi-day-to-weekly revisit cadence, and struggles with very large or very rapid deformation.

SENSE-MINE is positioned to fill this specific gap: a dense, continuous, low-cost surface sensing layer that produces high-frequency local deformation observations and immediate local alerts, complementing rather than replacing conventional surveying and InSAR.

1.2 Problem Statement (Reference)

Problem Statement ID: SIH26025 — "Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India." Organization: Ministry of Coal. Department: Coal India Limited.

The statement explicitly calls for: continuous monitoring of tilt, relative-distance change, crack initiation and vibration; AI-based anomaly detection and prediction; GIS-based risk-zone visualization; alerting; offline operation with synchronization; and scalability across sites.

1.3 Objectives

- Design and validate a low-cost, distributed, six-node surface sensor mesh capable of continuously measuring vertical displacement, crack opening, tilt and vibration over a simulated mining panel.
- Implement a self-healing multi-hop LoRa mesh so the network tolerates single-node communication failure without losing coverage.
- Build a three-layer AI pipeline (anomaly detection, deformation-state classification, short-horizon forecasting) that does not require a large labeled real-mine failure dataset to operate from day one.
- Fuse per-node AI outputs spatially and temporally so that a single noisy sensor cannot trigger a false mine-wide alert, while multi-node coherent movement is escalated with high confidence.
- Provide GIS-based risk visualization, independent offline and cellular alerting, and both offline (Node-RED) and online (web) dashboards.
- Demonstrate 24–72 hour continuous ("24×7-capable") operation before claiming production readiness.

2. Existing Technology and Positioning

2.1 Conventional Leveling / GNSS / Total Station

Very accurate at the point of measurement, and used as the reference standard for validating other methods. However, it is labor-intensive, requires periodic field campaigns rather than continuous monitoring, and is spatially sparse — it measures a small number of benchmark points, not a continuous field.

2.2 InSAR (Interferometric Synthetic Aperture Radar)

InSAR and SBAS-InSAR provide spatially continuous deformation monitoring over large areas and have been demonstrably applied to Indian coal-mine deformation, including work over coalfields in Chhattisgarh. Its limitations for this use case are structural rather than a matter of processing quality: satellite revisit cadence is days to weeks, measurement is line-of-sight only, and very large or very rapid deformation (beyond a few cm between passes) can decorrelate the signal entirely.

2.3 Wireless Sensor Networks (WSN) for Mine Monitoring

WSN-based approaches for underground and surface mine monitoring are an established research area, including real deployments for structural and environmental monitoring. Reliability, energy consumption over long deployments, and field maintainability remain the practical open challenges — which is exactly where SENSE-MINE's design choices (mesh routing with retries, solar/battery energy budgeting, sensor-health telemetry) are targeted.

2.4 SENSE-MINE's Positioning

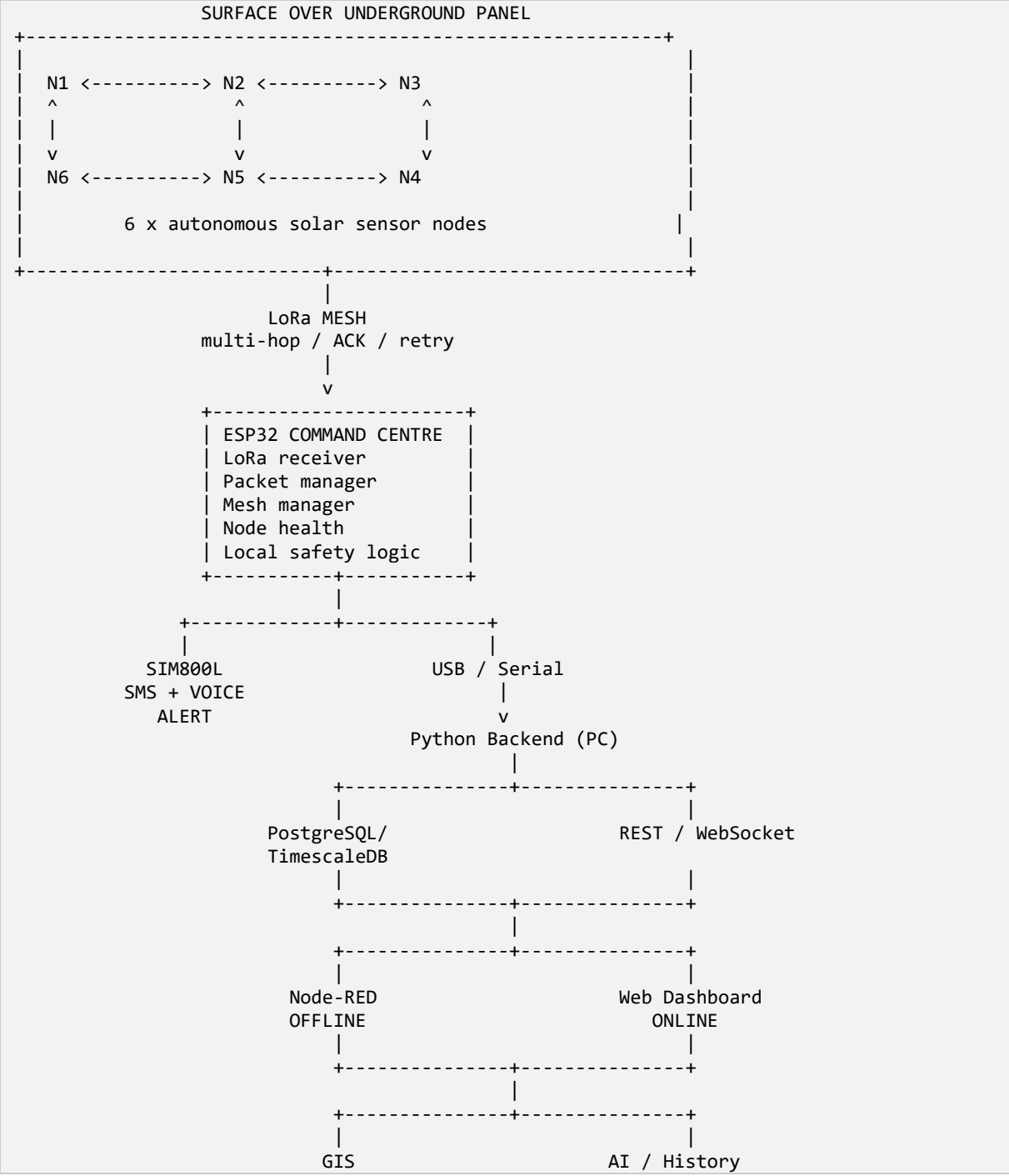
SENSE-MINE does not claim to invent subsidence monitoring, replace InSAR, or replace conventional surveying. Its innovation claim is narrower and more defensible:

> *"An integration of low-cost distributed surface deformation sensing, self-healing wireless mesh communication, multi-node spatial fusion, edge/central AI anomaly detection, GIS risk mapping, and independent offline/cellular alerting into a single low-cost architecture designed for continuous monitoring over Indian underground coal panels — positioned as a dense local ground-truth sensing layer that complements, not replaces, conventional surveying and InSAR."*

3. Proposed Solution — System Overview

SENSE-MINE consists of six autonomous, solar-powered surface sensor nodes arranged over a mining panel, communicating over a self-healing LoRa mesh to a command centre. The command centre performs local packet management, node-health tracking, and safety-critical alerting independent of internet connectivity (via SIM800L SMS/voice), while a connected operator PC/laptop runs the Python backend, database, AI inference pipeline, and both an offline (Node-RED) and online (web) dashboard.

3.1 High-Level Network Topology



This is a redundant-path mesh, not six independent point-to-point LoRa links: if a node cannot reach the command centre on its preferred route, it forwards through a neighbour instead. That property — not the presence of six radios — is what justifies calling the network a "mesh."

3.2 What This System Does Not Claim

- Does not claim real underground coal-mine field data was used for AI training in the current prototype stage.
- Does not claim commercial/survey-grade geotechnical displacement accuracy from a low-cost potentiometric transducer.
- Does not claim absolute long-horizon (24/48/72h) displacement forecasting until sufficient real field time-series exists.
- Does not claim GPS-survey-grade or satellite-grade GIS positioning.
- Does not claim the interpolated (IDW) risk surface is measured deformation everywhere — it is a spatial estimate between sensor points.
- Does not claim OTA (over-the-air) model deployment for the finals build — model updates are deployed via validated USB firmware flashing.

4. Hardware Architecture — Locked Design

4.1 Sensor Node (× 6)

Every node is built around an ESP32-S3, chosen for sufficient processing headroom for edge feature extraction and compact-model inference, a rich peripheral set (ADC, I²C, SPI), deep-sleep support for power budgeting, low unit cost, and a mature Arduino/ESP-IDF toolchain.

Component	Measurement / Function	Notes	Importance
ESP32-S3 DevKitC-1	Node controller	Edge feature extraction, deep sleep, ADC/I²C/SPI	★★★★★
WH20 linear potentiometer #1	Crack opening	Low-cost potentiometric displacement transducer with calibrated mechanical linkage — not a precision geotechnical sensor	★★★★★
WH20 linear potentiometer #2	Vertical displacement	Same caveat as above; individually calibrated per node	★★★★★
MPU6050	Tilt X/Y	Derived from gravity vector, fused with gyro via complementary/Kalman filter	★★★★★
MPU6050 (same chip)	Vibration RMS / peak / dominant frequency	Higher sampling rate, features computed on-node	★★★★
DS18B20	Temperature	Drift compensation for other sensors	★★★
GPS/GNSS module	Node location (installation-time only)	Fix obtained once at install, then powered off — not a continuous deformation sensor	★★★
SX1278 / RA-02 (433 MHz)	Mesh radio	HELLO / ROUTE / DATA / ACK / RETRY / HEARTBEAT protocol	★★★★★
Solar panel (5–6V, ~3W)	Energy harvesting	Sized against calculated node energy budget, not arbitrarily	★★★★
Li-ion battery pack (protected)	Energy storage	Solar charge/protection circuit, regulated 3.3V rail	★★★★
OLED + status LED (+ optional buzzer)	Local interface	On-site diagnostics without needing the dashboard	★★

4.2 Command Centre

Component	Function	Key Spec
ESP32-S3	LoRa mesh sink, packet manager, node health, local safety logic	Runs mesh + alert firmware; not a database or web server
SX1278 / RA-02	LoRa receiver from all 6 nodes	433 MHz
SIM800L	Independent SMS + voice-call alert channel	Quad-band; requires Airtel/Vi/BSNL SIM (not Jio — no 2G)
Dedicated 3.7–4.2V / ~2A regulator	Stable GSM power	Not drawn from the 3.3V logic rail

Component	Function	Key Spec
USB-Serial (permanent tether)	Link to operator PC	Carries packets to the Python backend; always-on, independent of Wi-Fi

Correction from an earlier design draft: the Python backend, database, AI inference and both dashboards run on the connected operator PC/laptop, not on the ESP32 command-centre board itself. The ESP32 command centre remains the LoRa gateway and safety-critical alert trigger; this keeps the architecture realistic and preserves offline capability even if the PC is temporarily unavailable, since local safety logic (CRITICAL → SIM800L) lives on the ESP32 itself.

4.3 Power Architecture

Per-node power is designed against an explicit energy budget rather than an arbitrarily chosen solar panel size:

```
Solar panel (5-6V, ~3W)
      |
Solar charge/protection circuit
      |
Protected Li-ion cell (>=3 Ah recommended)
      |
Low-quiescent-current regulator
      |
  +---+---+-------------+
  |       |             |
ESP32   SX1278      Switched rail
S3      (mesh)      (GPS / OLED / peripherals)
```

Energy-budget calculation to be finalized before purchasing the final BOM:

```
E_daily = P_avg x 24h,  P_avg = weighted average of: ESP32-S3 active/sleep current, LoRa
TX/RX current,
GPS current (installation-only, amortized), MPU6050, OLED, DS18B20, ADC, regulator losses,
deep-sleep current
```

Solar panel and battery capacity are then sized against E_daily with a worst-case (multi-cloudy-day) margin, not chosen because "the node is low power."

4.4 GPS Strategy

GPS is an installation/location-registration sensor, not a continuously-running deformation sensor: it obtains and validates a fix (checking HDOP) once at installation, stores latitude/longitude/elevation, and then powers off. Subsequent transmissions carry only the node ID and sensor data, referencing the stored coordinates — this removes GPS from the continuous power budget entirely.

```
Installation -> GPS obtains fix -> Validate fix/HDOP -> Store lat/lon/elevation -> GPS OFF
Runtime transmission: node_id + sensor_data (location resolved server-side from stored
coordinates)
```

5. Communication — LoRa Mesh Protocol

5.1 Protocol Frames

Rather than simple broadcast of raw sensor packets, the final firmware implements a discovery-and-routing protocol:

```
HELLO -> NEIGHBOUR_DISCOVERY -> ROUTE_UPDATE -> DATA -> ACK -> RETRY -> HEARTBEAT
```

Each node maintains a local routing table:

Field	Purpose
Neighbour ID	Identity of each node in radio range
RSSI / SNR	Link quality to that neighbour
Last seen	Recency of last successful exchange
Hop count	Distance (in hops) to the command centre via this neighbour
Route quality	Composite score used to select the best next hop

5.2 Route Recovery Example

```
Nominal:   N1 -> N2 -> N3 -> CC
N2 fails:  N1 -> N4 -> CC   (alternate route selected automatically)
```

Validation test: with the full mesh running, deliberately power off one intermediate node and confirm the network reroutes rather than losing coverage from the nodes behind it.

5.3 Sampling and Transmission Rates

Not every sensor runs at maximum rate continuously — this is the primary lever for meeting the 24×7 solar/battery energy budget.

Parameter	Suggested Rate
Crack displacement	1–5 Hz
Vertical displacement	1–5 Hz
Tilt	~10 Hz
Vibration	~100 Hz burst, processed locally to RMS/peak/dominant frequency before transmission
Temperature	~1 sample/min
GPS	Installation-time only, or periodic health check
LoRa transmission	Configurable, e.g. every 10–60 s
Heartbeat	Every 1–5 min

Only computed vibration features — not the raw 100 Hz stream — are transmitted over LoRa, keeping packets small and the radio/battery budget reasonable.

6. Sensor Node Measurement Pipeline and Calibration

6.1 On-Node Pipeline

```
PHYSICAL GROUND -> Sensors -> Raw ADC/I2C data -> Sensor calibration -> Noise filtering
-> Feature extraction -> Local anomaly screening -> LoRa mesh packet -> Command Centre
```

6.2 Calibration Layer

```
ADC -> Voltage -> Calibration curve -> Physical unit
e.g.  ADC -> crack width (mm)     ADC -> vertical displacement (mm)
```

Conversion coefficients come from a physical calibration experiment per node — the system explicitly does not assume a fixed scaling such as "4095 ADC counts = 100 mm."

6.3 Displacement / Crack-Opening Calibration Protocol

For each node, apply a series of known reference displacements (e.g. 0, 1, 2, 5, 10, 20, 30 mm) using a vernier or dial indicator, and record the corresponding ADC reading to build an individual ADC-to-physical-unit calibration curve. Compute calibration error per node:

```
Error(%) = |Sensor - Reference| / Reference x 100
```

This is performed separately for the crack-opening channel and the vertical-displacement channel on every one of the six nodes.

6.4 MPU6050 Calibration

- Accelerometer: six-position calibration (+X, –X, +Y, –Y, +Z, –Z) to estimate offset and scale factor.
- Gyroscope: several thousand stationary samples to compute gyro_bias_x/y/z, continuously subtracted at runtime.
- Tilt estimate: accelerometer-derived tilt fused with gyroscope angular rate via a complementary/Kalman filter for a stable estimate.

6.5 ADC Resolution vs. Accuracy

With a 12-bit ADC and a typical ~74 mm effective potentiometer travel, the ADC quantization step is approximately 0.018 mm/count. This is the ideal quantization resolution, not the actual measurement accuracy — real accuracy is bounded by potentiometer tolerance (typically ±20%) and mechanical linkage slop, and is established empirically per node via §6.3, not assumed from ADC bit depth.

7. Data Architecture

7.1 End-to-End Data Flow

```
ESP32-S3 (Command Centre) --USB Serial--> Python ingestion service (FastAPI) -->
PostgreSQL/TimescaleDB
ESP32-S3 (Command Centre) --USB Serial--> Node-RED --> Offline dashboard (independent of
the Python service)
```

Node-RED continues providing local monitoring even if the Python backend crashes — the two paths are deliberately independent.

7.2 Backend (FastAPI) Endpoints

```
/api/nodes      /api/readings    /api/history    /api/alerts
/api/risk       /api/gis         /api/health     /api/models
```

7.3 Database Schema

PostgreSQL with the TimescaleDB extension for time-series data:

Table	Contents
nodes	Node ID, install coordinates, hardware revision, calibration reference
sensor_readings	Raw per-node readings (crack width, displacement, tilt, vibration, temperature)
derived_features	Velocity, acceleration, rates, spatial gradient, spatial coherence
node_health	Battery, solar voltage, RSSI, SNR, packet loss, route/hop info, last-seen timestamp
alerts	Alert events, severity, channel, acknowledgement
predictions	XGBoost short-horizon forecast outputs
risk_history	Historical risk_level and deformation_state per node/panel
mesh_routes	Routing table snapshots for network diagnostics
calibration	Per-node, per-sensor calibration coefficients and calibration-date metadata

7.4 Example Record Shape

```
{
  "timestamp": "2026-09-09T17:10:25Z", "node_id": 3,
  "crack_width_mm": 4.21, "vertical_displacement_mm": 7.32,
  "tilt_x_deg": 0.42, "tilt_y_deg": -1.18,
  "vibration_rms_g": 0.034, "vibration_peak_g": 0.071, "dominant_frequency_hz": 4.88,
  "temperature_C": 31.2
}
Derived:
{
  "crack_growth_rate_mm_day": 0.83, "vertical_displacement_velocity_mm_day": 1.42,
  "displacement_acceleration_mm_day2": 0.21, "tilt_rate_deg_day": 0.16,
  "spatial_gradient": 3.42, "spatial_coherence": 0.91
}
AI output:
{ "anomaly_score": 0.87, "deformation_state": "ACCELERATING", "prediction_1h_mm": 9.4,
"risk_level": "ORANGE" }
```

7.5 Data Retention Strategy

Tier	Policy
0–90 days	Full resolution, live database
> 90 days	Hourly statistical aggregates
Permanent	Compressed raw archive — never deleted, retained for future model retraining

Data is never overwritten; every record is timestamped and appended, preserving a complete historical record for future AI retraining once real field events accumulate.

8. AI / ML Architecture

The AI pipeline uses three purpose-built layers rather than one model responsible for everything, so that each layer's claim is individually defensible.

8.1 Layer 1 — Isolation Forest (Anomaly Detection)

Purpose: detect unusual behaviour without requiring a large labeled real-mine-failure dataset — unsupervised anomaly detection is the defensible choice at this deployment stage, where genuinely labeled failure events do not yet exist.

Inputs: crack width, vertical displacement, tilt X/Y, vibration RMS, vibration peak, dominant frequency, crack growth rate, displacement velocity, displacement acceleration, tilt rate, spatial gradient, spatial coherence.

Output: NORMAL / ANOMALOUS / HIGHLY ANOMALOUS.

8.2 Layer 2 — Random Forest (Deformation-State Classification)

Output states: STABLE, INITIATION, PROGRESSIVE, ACCELERATING, RAPID, STABILIZING. These are presented as AI-derived deformation states — not official mine-safety categories — with the actual safety classification produced downstream by the risk engine (§9).

8.3 Layer 3 — XGBoost (Short-Horizon Forecasting)

Predicts near-term displacement using lagged features (D(t), D(t-1), D(t-2), D(t-5), velocity, acceleration, tilt, tilt rate, crack width, crack growth, vibration, spatial gradient, spatial coherence). Initial prediction horizons: 10 min, 30 min, 1 hour — extended later only once sufficient validated real field history exists. LSTM/Transformer models are explicitly deferred: the available dataset does not yet justify that complexity, and a simpler, auditable model is easier to defend in front of a technical jury.

8.4 Spatial Fusion (Core Innovation)

A single node's anomaly is not, by itself, sufficient evidence of subsidence. The system computes:

```
Spatial gradient:   G = max(D_i) - min(D_i)
Spatial coherence:  agreement of neighbouring-node trends over time
```

Example — coherent, high-confidence deformation:

```
N1 GREEN   N2 GREEN   N3 RED   N4 RED   N5 GREEN   N6 GREEN
=> N3/N4 form a spatially coherent abnormal region => HIGH CONFIDENCE DEFORMATION
```

Example — isolated, low-confidence reading:

```
N1 GREEN   N2 GREEN   N3 RED   N4 GREEN   N5 GREEN   N6 GREEN
=> single isolated reading => LOCAL ANOMALY / POSSIBLE SENSOR FAULT (not declared as mine-wide subsidence)
```

8.5 Fusion Engine (Full Pipeline)

```
                SENSOR DATA
                     |
              Sensor validation
                     |
        +------------+------------+
        |                         |
  Isolation Forest            RF classifier
        |                         |
        +------------+------------+
                     |
              Spatial fusion
                     |
              Temporal trend
                     |
```

```
                XGBoost forecast
                       |
                  RISK ENGINE
                       |
              GREEN / YELLOW / ORANGE / RED
```

8.6 Sensor-Fault Protection

The ML model never directly triggers an alert. A rule engine sits downstream of the AI outputs, cross-checking sensor health and spatial coherence before allowing an escalation — this is what prevents "AI says danger" from being the entire safety argument.

```
AI -> Anomaly score -> Spatial validation -> Sensor-health validation -> Rule engine ->
Risk level -> Alert
```

9. Risk Classification and Alert System

9.1 Risk Levels

Level	Meaning
GREEN	Normal
YELLOW	Early abnormality — single-node or low-confidence signal
ORANGE	Persistent / spatially coherent deformation across neighbouring nodes
RED	Rapid or accelerating deformation with multi-node confirmation

9.2 Alert Channels (Independent, Redundant)

Channel	Mechanism	Works Offline?
Local	OLED display + buzzer at the node/command centre	Yes
Cellular	SIM800L SMS + voice call (rings only, no spoken message)	Yes
Online	Browser alert / push / email via the web dashboard	Requires internet

Design principle: the safety alert path must not depend exclusively on internet connectivity — the SIM800L cellular path at the command centre provides an independent channel, and requires an Airtel/Vi/BSNL SIM since Jio has no 2G network.

9.3 Alert Hysteresis

Escalation and SMS/call triggers use hysteresis/cooldown logic so that sensor noise near a threshold does not repeatedly re-trigger alerts — an alert fires on a sustained state change, not a single noisy sample.

9.4 Node-Health vs. Subsidence Distinction

A missed heartbeat is reported as a communication failure, never conflated with a subsidence event:

```
NODE 3   Last packet: 18:21:31   Current: 18:21:36   Status: ONLINE
If no packet beyond threshold -> "NODE COMMUNICATION FAILURE" (not "SUBSIDENCE")
```

10. GIS and Visualization

10.1 Map Layers

- Mine boundary and panel boundary
- Underground panel / goaf geometry (reference overlay)
- N1–N6 real GPS-registered sensor locations
- Live per-node status (GREEN/YELLOW/ORANGE/RED)
- Deformation vectors
- Interpolated risk heatmap (IDW — Inverse Distance Weighting)
- Historical playback of past risk states

10.2 Implementation

Web GIS built on Leaflet + OpenStreetMap tiles, with sensor locations pinned from installation-time GPS fixes (§4.4). Between sensor points, an IDW-interpolated field is generated and is explicitly labeled "Estimated spatial risk/deformation field," not "actual deformation everywhere" — this distinction is maintained on the UI itself, not just in documentation.

11. Dashboards — Offline and Online

11.1 Offline Dashboard (Node-RED)

Page	Contents
Live Monitor	Per-node status lights + overall panel risk level
Sensor Data	Time-series charts: crack width, vertical displacement, tilt X/Y, vibration RMS/peak, dominant frequency, temperature, velocity, acceleration
GIS	Live sensor map
AI	Anomaly score, deformation state, predicted displacement, prediction confidence, risk level
Node Health	Battery, solar, RSSI, SNR, packet loss, last seen, route, sensor status
Alert History	Time, node, event, severity, AI state, action, acknowledgement

11.2 Online Dashboard (React + Web)

Mirrors the offline dashboard's information with the additions needed for a production monitoring platform: long-term history, remote access, historical playback, cross-node comparison, AI predictions, alert management, and data export/reporting.

12. System Reliability and 24×7 Operation

12.1 Node Heartbeat and Health Telemetry

Every node reports battery voltage, solar voltage, RSSI, SNR, packet loss, and current mesh route alongside sensor data — these are treated as first-class data fields for a 24×7 system, not optional dashboard extras (§7.3).

12.2 Firmware Correctness Prerequisites

Before the final firmware is frozen, the following are corrected and validated — prioritized above adding further AI models, since a safety system built on bad rate math is a worse risk than a missing feature:

- Timestamp / timebase consistency across all nodes
- Rate calculations (crack growth rate, tilt rate) — a known prior issue produced physically impossible values (thousands of mm/day) from a time-unit error, and must not reach the AI/risk engine
- Per-node calibration coefficients (§6.3–§6.4)
- NaN handling for missing/invalid sensor reads (a known prior issue: temperature_C = NaN)
- Sensor-health flags
- Packet sequence numbers
- Battery-voltage reporting
- Heartbeat reliability
- Mesh routing correctness (route recovery test, §5.2)

12.3 Endurance Testing

The system is not described as "24×7 capable" until an actual continuous endurance test (recommended: 24–72 hours, all six nodes running simultaneously) has been run and its results — sensor uptime, ADC stability, LoRa packet delivery rate, battery/solar behaviour, memory usage, and any resets — are recorded.

13. Implementation Plan

Build order is deliberately hardware-and-reliability-first, not dashboard-first:

Phase	Deliverable
1	One production-quality node (ESP32-S3 + sensors + LoRa + battery + solar + enclosure) run continuously; validate sensor uptime, ADC/MPU stability, LoRa delivery, power draw
2	Replicate to all six nodes
3	Implement mesh protocol: HELLO / ROUTE / DATA / ACK / RETRY / HEARTBEAT; test route recovery by deliberately disabling one node
4	Calibrate every sensor on every node; generate and store calibration coefficients
5	Command centre firmware: LoRa mesh sink, packet validation, node health, SIM800L integration, USB link
6	Python backend: Serial ingestion -> FastAPI -> PostgreSQL/TimescaleDB
7	AI layer: Isolation Forest, Random Forest, XGBoost, trained/validated on clearly-labeled demonstration data
8	Risk engine: AI outputs + spatial coherence + temporal acceleration + sensor health -> risk level
9	Node-RED offline dashboard
10	Online web dashboard
11	GIS integration (Leaflet + IDW risk surface)
12	24–72 hour endurance test across all six nodes — only after this is the system described as 24×7-capable

Deliberately deferred out of scope for the finals build: OTA (over-the-air) model deployment. The current model is compiled directly into firmware (e.g. model_rf_compact.cpp); OTA would require substantially more firmware infrastructure and risks destabilizing the demonstration build. Model updates instead follow: historical data → training pipeline → validation → human approval → model generation → firmware compilation → USB deployment.

14. Testing and Validation Plan

Test	Method	Pass Criterion
Displacement/crack calibration accuracy	Known reference displacements (vernier/dial indicator) vs. sensor reading, per node	Error(%) within an agreed tolerance band, documented per node
MPU6050 calibration	Six-position accelerometer test + stationary gyro-bias collection	Offset/scale/bias coefficients stored and applied at runtime
Mesh route recovery	Disable an intermediate node mid-run	Downstream nodes reroute and continue reporting
Sensor-fault discrimination	Simulate a single-node stuck/jump/drift fault with neighbours held stable	System reports LOCAL ANOMALY / POSSIBLE SENSOR FAULT, not panel-wide RED
Spatial-coherence escalation	Simulate coordinated multi-node displacement (N3/N4 rising together)	System reports HIGH CONFIDENCE DEFORMATION / escalates to ORANGE–RED
Alert hysteresis	Oscillate a reading near a threshold	No repeated SMS/call spam; single sustained-state alert
Offline resilience	Disconnect internet from the operator PC	Node-RED dashboard and local alerting continue functioning
24–72h endurance	Continuous run, all six nodes	No unrecovered resets; heartbeat/packet-delivery rate logged and reported

15. Cost Estimation

Indicative per-node and command-centre bill of materials (approximate INR, prototype-quantity pricing):

Item	Approx. Unit Cost (INR)	Qty (per node)	Notes
ESP32-S3 DevKitC-1	700–900	1	Also used as command-centre MCU
WH20 linear potentiometer (x2)	80–150 each	2	Crack opening + vertical displacement
MPU6050	90–150	1	Tilt + vibration
DS18B20	80–150	1	Temperature
GPS/GNSS module	250–600	1	Installation-time fix only
SX1278 / RA-02	400–500	1	Mesh radio
Solar panel (~3W)	250–400	1	Sized against energy budget, §4.3
Li-ion cell + protection/charge circuit	300–500	1	
OLED + LED + buzzer	150–250	1	Local interface
Enclosure + mounting hardware	200–400	1	Field-deployable housing

Command centre adds: ESP32-S3, SX1278, SIM800L (₹184–400), dedicated GSM regulator (₹150–250), USB cable (~₹50). Total system cost (6 nodes + command centre) is estimated in the low tens of thousands of INR at prototype quantities — the exact figure is finalized once the energy-budget calculation (§4.3) fixes the final solar/battery sizing.

16. Innovation and Unique Selling Points

- Self-healing multi-hop LoRa mesh with ACK/retry/route-recovery, rather than independent point-to-point radios.
- Multi-node spatial fusion (gradient + coherence) as a first-class part of the risk decision, distinguishing genuine coherent deformation from an isolated faulty sensor before any alert is raised.
- A three-layer, explainable AI pipeline that does not require a large labeled real-mine-failure dataset to be useful from the first deployment.
- A rule engine downstream of the AI outputs, so no single model output can unilaterally trigger a safety alert.
- Independent offline (Node-RED, local DB) and cellular (SIM800L) paths, so core monitoring and alerting do not depend on internet connectivity.
- Honest scientific framing throughout: potentiometric transducers are labeled as low-cost transducers with calibrated linkage, not precision geotechnical instruments; forecasts are short-horizon trend classifications, not absolute multi-hour predictions; the IDW surface is labeled an estimate, not a measurement.

17. Feasibility and Viability

17.1 Technical Feasibility

All components are commercially available, low-cost, and use mature, well-documented interfaces (I^2C, SPI, 1-Wire, analog ADC). The AI layers (Isolation Forest, Random Forest, XGBoost) are lightweight enough to run on a standard operator PC without specialized hardware, and edge feature extraction fits comfortably within the ESP32-S3's compute/memory budget.

17.2 Operational Feasibility

Solar + battery power, mesh route recovery, and independent cellular alerting are specifically chosen to support field deployment away from reliable grid power or internet — conditions typical of surface areas over underground panels.

17.3 Economic Feasibility

Total hardware cost (§15) is a small fraction of the cost of comparable survey-grade instrumentation or an InSAR service contract, positioning SENSE-MINE as a continuous supplementary layer rather than a capital-intensive replacement for existing monitoring.

18. Social and Safety Impact

- Continuous, near-real-time local monitoring can surface early warning signs between periodic survey campaigns, when subsidence risk to workers and nearby structures may otherwise go undetected for days or weeks.
- Low unit cost supports denser sensor placement than survey-grade instrumentation typically allows, improving spatial coverage over specific panels of concern.
- Independent offline and cellular alerting keeps the safety function available even where site connectivity is unreliable — a realistic condition at many mine surface sites.

19. Known Limitations (Stated Explicitly)

- This is a scaled physical/prototype validation platform, not a geomechanical replica of an underground coal mine.
- No official/field dataset exists for PS 26025 at this stage; AI training/demonstration uses clearly labeled synthetic and testbed data, not real mine failure records.
- Potentiometric displacement sensors are low-cost transducers with calibrated mechanical linkage, not survey-grade geotechnical instruments — accuracy is bounded by component tolerance (±20% typical) and established per-node via calibration, not assumed.
- Vibration and tilt are derived from the same physical sensor (MPU6050); this is documented, not concealed.
- Short-horizon (10 min–1 hour) trend forecasting only; multi-hour/day-scale absolute forecasting is not claimed until validated against sufficient real field time-series.
- GPS is used for installation-time location registration, not continuous positioning; GIS is a schematic/Leaflet overlay with surveyed local coordinates, not satellite-grade GIS.
- The IDW-interpolated risk surface is an estimate between measurement points, not a claim of measured deformation everywhere.
- SIM800L requires a 2G-capable SIM (Airtel/Vi/BSNL); Jio SIMs will not work, and 2G coverage is a shrinking national resource — flagged as a future-upgrade path, not hidden.
- OTA model deployment is out of scope for the finals build; models are validated and deployed via USB firmware flashing.
- The system has not yet undergone field validation against a real underground coal mine or an independent survey benchmark — flagged as a scope limit for this stage, with physical testbed validation and a defined endurance-test target as the interim substitute (§12.3, §14).
- Self-calibrating baseline concepts and mesh networking are approaches multiple teams addressing this problem statement are likely to converge on; SENSE-MINE's differentiation rests on spatial fusion, the explainable rule-engine risk decision, and the honestly-scoped data strategy — not on either of those individually.

20. Future Scope

Upgrade	What It Adds	Why Deferred
Independent geophone (SM-24) per node	True independent vibration channel	Cost and stocking availability in India
Piezometer / pore-water-pressure sensing	A real leading indicator currently unmeasured	Budget-constrained at prototype stage
GNSS-RTK reference stations	±1–2 cm real-time field validation	Requires base + rover hardware beyond current budget
InSAR/SBAS integration	Km-scale, mm-precision context for a real mine	Requires SAR processing pipeline + institutional partnership
4G/LTE module (e.g. SIM7600) vs. 2G SIM800L	Removes dependency on shrinking 2G coverage	Higher cost/complexity than SIM800L for the prototype
True multi-hop relay at larger scale	Extends mesh coverage beyond a 6-node demonstration	6-node star/mesh sufficient to demonstrate the concept
Offline→online data sync	Fully satisfies the PS's literal offline+sync wording	Deliberately scoped out for finals; straightforward to add later
OTA model deployment	Remote model updates without physical access	Deferred to avoid destabilizing the finals demonstration build
Field validation with a mine/research institution	Real ground-truth validation vs. survey benchmarks	Requires institutional access beyond current scope

21. Conclusion

SENSE-MINE integrates a low-cost, solar-powered, self-healing six-node LoRa mesh with a three-layer, explainable AI pipeline and a spatial-fusion risk engine to deliver continuous local surface-deformation monitoring over underground coal panels. Its design deliberately avoids overclaiming: sensor accuracy, forecasting horizon, GIS precision, and dataset provenance are each stated at the level the current prototype can actually defend, while the architecture, database schema, and phased implementation plan are built to extend cleanly toward field validation and eventual production deployment. Positioned as a dense, continuous, local complement to conventional surveying and InSAR — not a replacement for either — SENSE-MINE directly addresses the core asks of Problem Statement 26025: low-cost distributed sensing, mesh networking, AI-based anomaly detection and prediction, GIS risk visualization, independent alerting, and offline-capable, scalable operation.

22. References

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